Ex. **(2)** The first term a and common difference d are given. Find first four terms of A.P.

(i) $a = -3, d = 4$ (ii) $a = 200, d = 7$

(iii) $a = -1, d = -\frac{1}{2}$ (iv) $a = 8, d = -5$

Solution : (i) Given $a = -3, d = 4$

$t_1 = -3$

$t_2 = t_1 + d = -3 + 4 = 1$

$t_3 = t_2 + d = 1 + 4 = 5$

$t_4 = t_3 + d = 5 + 4 = 9$

$\therefore$ A.P. is $= -3, 1, 5, 9, \ldots$

(ii) Given $a = 200, d = 7$

$a = t_1 = 200$

$t_2 = t_1 + d = 200 + 7 = 207$

$t_3 = t_2 + d = 207 + 7 = 214$

$t_4 = t_3 + d = 214 + 7 = 221$

$\therefore$ A.P. is $= 200, 207, 214, 221, \ldots$

(iii) $a = -1, d = -\frac{1}{2}$

$a = t_1 = -1$

$t_2 = t_1 + d = -1 + (-\frac{1}{2}) = -\frac{3}{2}$

$t_3 = t_2 + d = -\frac{3}{2} + (-\frac{1}{2}) = -\frac{4}{2} = -2$

$t_4 = t_3 + d = -2 + (-\frac{1}{2})$

$= -2 - \frac{1}{2} = -\frac{5}{2}$

$\therefore$ A.P. is $= -1, -\frac{3}{2}, -2, -\frac{5}{2}, \ldots$

(iv) $a = 8, d = -5$

$a = t_1 = 8$

$t_2 = t_1 + d = 8 + (-5) = 3$

$t_3 = t_2 + d = 3 + (-5) = -2$

$t_4 = t_3 + d = -2 + (-5) = -7$

$8, 3, -2, -7, \ldots$

$\therefore$ A.P. is $= 8, 3, -2, -7, \ldots$

Practice Set 3.1

1. Which of the following sequences are A.P. ? If they are A.P. find the common difference .

(1) $2, 4, 6, 8, \ldots$ (2) $2, \frac{5}{2}, 3, \frac{7}{3}, \ldots$ (3) $-10, -6, -2, 2, \ldots$

(4) $0.3, 0.33, .0333, \ldots$ (5) $0, -4, -8, -12, \ldots$ (6) $-\frac{1}{5}, -\frac{1}{5}, -\frac{1}{5}, \ldots$

(7) $3, 3 + \sqrt{2}, 3 + 2\sqrt{2}, 3 + 3\sqrt{2}, \ldots$ (8) $127, 132, 137, \ldots$

2. Write an A.P. whose first term is a and common difference is d in each of the following.

(1) $a = 10, d = 5$ (2) $a = -3, d = 0$ (3) $a = -7, d = \frac{1}{2}$

(4) $a = -1.25, d = 3$ (5) $a = 6, d = -3$ (6) $a = -19, d = -4$

3. Find the first term and common difference for each of the A.P.

(1) 5, 1, −3, −7, . . . (2) 0.6, 0.9, 1.2, 1.5, . . .

(3) 127, 135, 143, 151, . . . (4) $\frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}, \ldots$



Let's think.

- Is 5, 8, 11, 14, . . . an A.P. ? If so then what will be the 100th term? Check whether 92 is in this A.P.? Is number 61 in this A.P.?



Let's learn.

n^{th} term of an A. P.

In the sequence 5, 8, 11, 14, . . . the difference between two consecutive terms is 3. Hence, this sequence is an A.P.

Here the first term is 5. If 3 is added to 5 we get the second term 8. Similarly to find 100th term what should be done?

	First term	Second term	Third term	. . .
Number	5,	5 + 3 = 8	8 + 3 = 11	. . .

In this way reaching upto 100th term will be time consuming. Let's see if we can find any formula for it.

5	8	11	14	. . .	. . .	. . .	. . .
5	$5 + 1 \times 3$	$5 + 2 \times 3$	$5 + 3 \times 3$	. . .	$5 + (n - 1) \times 3$	$5 + n \times 3$	. . .
1st term t_1	2nd term t_2	3rd term t_3	4th term t_4	. . .	n^{th} term t_n	$(n + 1)^{th}$ term t_{n+1}	. . .

Generally in the A.P. $t_1, t_2, t_3, \ldots$ If first term is a and common difference is d,

$t_1 = a$

$t_2 = t_1 + d = a + d = a + (2 - 1)\, d$

$t_3 = t_2 + d = a + d + d = a + 2d = a + (3 - 1)d$

$t_4 = t_3 + d = a + 2d + d = a + 3d = a + (4 - 1)d$

We get $t_n = a + (n - 1)\, d.$

Using the above formula we can find the 100th term of the A.P. 5, 8, 11, 14, . . .

Here $a = 5 \quad d = 3$

$$t_n = a + (n - 1)d$$

$$\therefore \quad t_{100} = 5 + (100 - 1) \times 3$$

$$= 5 + 99 \times 3$$

$$= 5 + 297$$

$$t_{100} = 302$$

100th tem of this A.P. is 302.

Let's check whether 61 is in this A.P. To find the answer we use the same formula.

$$t_n = a + (n - 1)d$$

$$t_n = 5 + (n - 1) \times 3$$

If 61 is n^{th} term means t_n, then

$$\therefore \quad 61 = 5 + 3n - 3$$

$$= 3n + 2$$

$$\therefore \quad 3n = 59$$

$$\therefore \quad n = \frac{59}{3}$$

But then, n is not a natural number.

$\therefore$ 61 is not in this A.P.



Let's think.

Kabir's mother keeps a record of his height on each birthday. When he was one year old, his height was 70 cm, at 2 years he was 80 cm tall and 3 years he was 90 cm tall. His aunt Meera was studying in the 10th class. She said, ''it seems like Kabir's height grows in Arithmetic Progression''. Assuming this, she calculated how tall Kabir will be at the age of 15 years when he is in the 10th ! She was shocked to find it. You too assume that Kabir grows in A.P. and find out his height at the age of 15 years.

Solved examples

Ex. (1) Find t_n for following A.P. and then find 30th term of A.P.

3, 8, 13, 18, . . .

Solution :Given A.P. 3, 8, 13, 18, . . .

Here $t_1 = 3,\ t_2 = 8, t_3 = 13, t_4 = 18, \ldots$

$d = t_2 - t_1 = 8 - 3 = 5$

We know that $t_n = a + (n - 1)d$

$\therefore\ t_n = 3 + (n - 1) \times 5 \quad \because\ a = 3,\ d = 5$

$\therefore\ t_n = 3 + 5n - 5$

$\therefore\ t_n = 5n - 2$

$\therefore\ 30^{th}$ term $= t_{30} = 5 \times 30 - 2$

$= 150 - 2 \quad = 148$

Ex. (2) Which term of the following A.P. is 560?

2, 11, 20, 29, . . .

Solution : Given A.P. 2, 11, 20, 29, . .

Here $a = 2, d = 11 - 2 = 9$

n^{th} term of this A.P. is 560.

$t_n = a + (n - 1)d$

$\therefore\ 560 = 2 + (n - 1) \times 9$

$= 2 + 9n - 9$

$\therefore\ 9n = 567$

$\therefore\ n = \frac{567}{9} = 63$

$\therefore\ 63^{rd}$ term of given A.P. is 560.

Ex. (3) Check whether 301 is in the sequence 5, 11, 17, 23, . . . ?

Solution : In the sequence 5, 11, 17, 23,. . .

$t_1 = 5,\ t_2 = 11, t_3 = 17, t_4 = 23, \ldots$

$t_2 - t_1 = 11 - 5 = 6$

$t_3 - t_2 = 17 - 11 = 6$

$\therefore$ This sequence is an A.P.

First term $a = 5$ and $d = 6$

If 301 is n^{th} term, then.

$t_n = a + (n - 1)d = 301$

$\therefore\ 301 = 5 + (n - 1) \times 6$

$= 5 + 6n - 6$

$\therefore\ 6n = 301 + 1 = 302$

$\therefore\ n = \frac{302}{6}$. But it is not an integer.

$\therefore$ 301 is not in the given sequence.

Ex. (4) How many two digit numbers are divisible by 4?

Solution : List of two digit numbers divisible by 4 is

12, 16, 20, 24, . . . , 96.

Let's find how many such numbers are there.

$t_n = 96, \quad a = 12, \quad d = 4$

From this we will find the value of n.

$t_n = 96$, $\therefore$ By formula,

$96 = 12 + (n - 1) \times 4$

$= 12 + 4n - 4$

$\therefore\ 4n = 88$

$\therefore\ n = 22$

$\therefore$ There are 22 two digit numbers divisible by 4.

Ex. (5) – The 10th term and the 18th term of an A.P. are 25 and 41 respectively then find 38th term of that A.P., similarly if n^{th} term is 99. Find the value of n.

Solution : In the given A.P. $t_{10} = 25$ and $t_{18} = 41$.

We know that, $t_n = a + (n - 1)d$

$\therefore \quad t_{10} = a + (10 - 1)\, d$

$\therefore \quad 25 = a + 9d \ldots$ (I)

Similarly $t_{18} = a + (18 - 1)\, d$

$\therefore \quad 41 = a + 17d \ldots$ (II)

$25 = a + 9d \ldots$ From (I) .

$a = 25 - 9d.$

Substituting this value in equation II.

$\therefore$ Equation (II) $a + 17d = 41$

$\therefore \quad 25 - 9d + 17d = 41$

$8d = 41 - 25 = 16$

$d = 2$

Substituting $d = 2$ in equation I.

$a + 9d = 25$

$\therefore\ a + 9 \times 2 = 25$

$\therefore\ a + 18 = 25$

$\therefore\ a = 7$

Now, $t_n = a + (n - 1)d$

$\therefore \quad t_{38} = 7 + (38 - 1) \times 2$

$t_{38} = 7 + 37 \times 2$

$t_{38} = 7 + 74$

$t_{38} = 81$

If n^{th} term is 99, then to find value of n.

$t_n = a + (n - 1)d$

$99 = 7 + (n - 1) \times 2$

$99 = 7 + 2n - 2$

$99 = 5 + 2n$

$\therefore\ 2n = 94$

$\therefore\ n = 47$

$\therefore$ In the given progression 38th term is 81 and 99 is the 47th term.

Practice Set 3.2

1. Write the correct number in the given boxes from the following A. P.

(i) 1, 8, 15, 22, . . .

Here $a =$ ☐, $t_1 =$ ☐, $t_2 =$ ☐, $t_3 =$ ☐,

$t_2 - t_1 =$ ☐ − ☐ = ☐

$t_3 - t_2 =$ ☐ − ☐ = ☐ $\therefore d =$ ☐

(ii) 3, 6, 9, 12, . . .

Here $t_1 =$ ☐, $t_2 =$ ☐, $t_3 =$ ☐, $t_4 =$ ☐,

$t_2 - t_1 =$ ☐, $t_3 - t_2 =$ ☐ $\therefore d =$ ☐

(iii) −3, −8, −13, −18, . . .

Here $t_3 =$ ☐, $t_2 =$ ☐, $t_4 =$ ☐, $t_1 =$ ☐,

$t_2 - t_1 =$ ☐, $t_3 - t_2 =$ ☐ $\therefore a =$ ☐, $d =$ ☐

(iv) 70, 60, 50, 40, . . .

Here $t_1 =$ ☐, $t_2 =$ ☐, $t_3 =$ ☐, . . .

$\therefore a =$ ☐, $d =$ ☐

2. Decide whether following sequence is an A.P., if so find the 20th term of the progression.

−12, −5, 2, 9, 16, 23, 30, . . .

3. Given Arithmetic Progression 12, 16, 20, 24, . . . Find the 24th term of this progression.

4. Find the 19th term of the following A.P.

7, 13, 19, 25, . . .

5. Find the 27th term of the following A.P.

9, 4, −1, −6, −11, . . .

6. Find how many three digit natural numbers are divisible by 5.

7. The 11th term and the 21st term of an A.P. are 16 and 29 respectively, then find the 41th term of that A.P.

8. 11, 8, 5, 2, . . . In this A.P. which term is number −151?

9. In the natural numbers from 10 to 250, how many are divisible by 4?

10. In an A.P. 17th term is 7 more than its 10th term. Find the common difference.

The Wise Teacher

Once upon a time, there lived a king. He appointed two teachers Tara and Meera to teach horse riding for one year to his children Yashwantraje and Geetadevi. He asked both of them how much salary they wanted.

Tara said, ''Give me 100 gold coins in first month and every month increase the amount by 100 gold coins.'' Meera said, ''Give me 10 gold coins in the first month and every month just double the amount of the previous month.''

The king agreed. After three months Yashwantraje said to his sister, ''My teacher is smarter than your teacher as she had asked for more money.'' Geetadevi said, ''I also thought the same, I asked Meeratai about it. She only smiled and said compare the salaries after 8 months. I calculated their 9 months salaries. You can also check.''

Months	1	2	3	4	5	6	7	8	9	10	11	12
Tara's salary	100	200	300	400	500	600	700	800	900	–	–	–
Meera's salary	10	20	40	80	160	320	640	1280	2560	–	–	–

Complete the above table.

Tara's salary 100, 200, 300, 400, . . . is in A.P.

$t_1 = 100$, $t_2 = 200$, $t_3 = 300$,. . . $t_2 - t_1 = 100 = d$

Common difference is 100.

Meera's salary 10, 20, 40, 80, . . . is not an A.P. Reason is $20 - 10 = 10$, $40 - 20 = 20$, $80 - 40 = 40$ So the common difference is not constant.

But here each term is double the preceding term.

Here $\frac{t_2}{t_1} = \frac{20}{10} = 2$, $\frac{t_3}{t_2} = \frac{40}{20} = 2$, $\frac{t_4}{t_3} = \frac{80}{40} = 2$

$\therefore \frac{t_{n+1}}{t_n}$ The ratio of a term and its preceding term is constant. This type of progression is called a geometric progression. Notice that if ratio $\frac{t_{n+1}}{t_n}$ is greater than 1, then geometric progression will increase faster than arithmetic progression.

If the ratio is smaller than 1, note how the geometric progression changes.

This year we are going to study Arithmetic Progression only. We have seen how to find the nth term of an A.P. Now we are going to see how to find the sum of the first n terms.

Quick Addition

Three hundred years ago there was a single teacher school in Germany. The teacher was Buttner and he had an assistant Johann Martin Bortels. He used to teach alphabets to the children and sharpen their pencils. Buttner was a strict teacher. One day he wanted to do some work and wanted peace in the class, so he tried to occupy all students with a lengthy addition. They were asked to add all intergers from 1 to 100. In few minutes one slate was slammed on the floor. He looked at Carl Gauss and asked, ''I asked you to add all integers from 1 to 100. Why did you keep the slate down? Don't you want to do it ?''

Carl Gauss said, ''I have done the addition.''

The teacher asked, ''How did you do it so quickly? You wouldn't have written all the numbers ! What is the answer ?''

Carl Gauss said, ''Five thousand fifty''

Teacher was so surprised and asked him, 'How do you find the answer?''

Carl Gauss explained his quick addition method:

Nos. in increasing order	1	2	3	4	. . . . .	100
	)+	)+	)+	)+		)+
Nos. in decreasing order	100	99	98	97	. . . . .	1
Sum	101	101	101	101		101

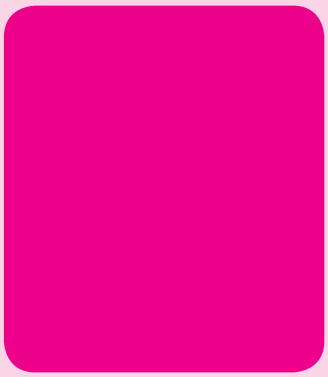
The sum of each pair is 101. This sum occurs 100 times so 101×100 is the product needed. It is 10100. In this 1 to 100 are counted two times. Therefore, half of 10100 is 5050 and sum of 1 to 100 is 5050. The teacher appreciated his work.

Now using this method of Gauss, let's find sum of n terms of an A.P.

Johann Friedrich Carl Gauss

30th April 1777 – 23rd February 1855.

Carl Gauss was a great German mathematician, He was born in Braunschweig, he was the only son of uneducated parents. He showed a glimpse of his intelligence in Buttner's school. After some years, Buttner's helper, Johann Martin Bartels and Gauss became friends. Together, they published a book on Algebra. Bartels made the other people realise the extra ordinary intelligence of Gauss.





Let's learn.

Sum of first n terms of an A. P.

Arithmetic Progression $a, a + d, a + 2d, a + 3d, \ldots\ a + (n - 1)d$

In this progression a is the first term and d is the common difference. Let's write the sum of first n terms as S_n.

$S_n = [a] + [a + d] + \ldots + [a+(n-2)d] + [a+(n-1)d]$

Reversing the terms and rewritting the expression again,

$S_n = [a+(n-1)d] + [a+(n-2)d] + \ldots + [a + d] + [a]$

On adding,

$2S_n = [a+a+(n-1)d] + [a + d+a+(n-2)d] + \ldots + [a+(n-2)d+ a + d] + [a+(n-1)d+a]$

$2S_n = [2a+(n-1)d] + [2a+(n-1)d] + \ldots + [2a+(n-1)d]$ . . . n times.

$\therefore\ 2S_n = n\,[2a+(n-1)d]$

$\therefore\ S_n = \frac{n}{2}\,[2a+(n-1)d]$ or $S_n = na + \frac{n(n-1)}{2}d$

Ex. Let's find the sum of first 100 terms of A.P. 14, 16, 18, ...

Here $a = 14$, $d = 2$, $n = 100$

$$S_n = \frac{n}{2}\,[2a+(n-1)d]$$

$$\therefore\ S_{100} = \frac{100}{2}\,[2 \times 14+(100-1) \times 2]$$

$$= 50\,[28 + 198]$$

$$= 50 \times 226 = 11300$$

$\therefore$ Sum of first 100 terms of given A.P. is 11,300



Let's remember!

For the given Arithmetic Progression, if first term is a and common diffrence is d then

$$t_n = [a+(n-1)d]$$

$$S_n = \frac{n}{2}\,[2a+(n-1)d] = na + \frac{n(n-1)}{2}d$$

Let's find one more formula for sum of first n terms.

In the A.P. $a, a+d, a+2d, a+3d, \ldots\ a+(n-1)d$

First term = $t_1 = a$ and n^{th} term is $[a+(n-1)d]$

Now $S_n = \frac{n}{2}\left[\underline{a}+\underline{a+(n-1)d}\right]$

$\therefore\ S_n = \frac{n}{2}[t_1 + t_n] = \frac{n}{2}$ [First term+ last term]

Solved examples

Ex. (1) Find the sum of first n natural numbers.

Solution : First n natural numbers are $1, 2, 3, \ldots, n$.

Here $a = 1,\ d = 1,\ n^{th}$ term $= n$

$\therefore\ S_n = 1 + 2 + 3 + \ldots + n$

$S_n = \frac{n}{2}$ [First term+ last term] . . . . . (by the formula)

$= \frac{n}{2}[1 + n]$

$= \frac{n(n+1)}{2}$

$\therefore$ Sum of first n natural number is $\frac{n(n+1)}{2}$.

Ex. (2) Find the sum of first n even natural numbers.

Solution : First n even natural numbers are $2, 4, 6, 8, \ldots, 2n$.

t_1 = First term = 2 , t_n = last term = $2n$

Method I

$= \frac{n}{2}[t_1 + t_n]$

$= \frac{n}{2}[2 + 2n]$

$= \frac{n}{2} \times 2\ (1+ n)$

$= n\ (1+ n)$

Method II

$S_n = 2 + 4 + 6 \ldots + 2n$

$= 2(1 + 2 + 3 + \ldots + n)$

$= \frac{2[n(n+1)]}{2}$

$= n\ (1+ n)$

Method III

$S_n = \frac{n}{2}[2a+(n-1)d]$

$= \frac{n}{2}[2 \times 2+(n-1)2]$

$= \frac{n}{2}[4+2n-2]$

$= \frac{n}{2}[2 + 2n]$

$= \frac{n}{2} \times 2\ (1+ n)$

$= n\ (1+ n)$

$\therefore$ Sum of first n even natural numbers is $n\ (1+ n)$.

Ex. (3) Find the sum of first n odd natural numbers.

Solution : First n natural numbers

$1, 3, 5, 7, \ldots, (2n - 1)$.

$a = t_1 = 1$ and $t_n = (2n - 1)$, $d = 2$

Method I

$S_n = \frac{n}{2}[t_1 + t_n]$

$= \frac{n}{2}[1 + (2n - 1)]$

$= \frac{n}{2}[1 + 2n - 1]$

$= \frac{n}{2} \times 2n$

$= n^2$

Method II

$S_n = \frac{n}{2}[2a + (n-1)d]$

$= \frac{n}{2}[2 \times 1 + (n-1) \times 2]$

$= \frac{n}{2}[2 + 2n - 2]$

$= \frac{n}{2} \times 2n$

$= n^2$

Method III

$1 + 3 + \ldots 2n-1$

$= (1 + 2 + 3 + \ldots + 2n)$

$- (2 + 4 + 6 + \ldots + 2n)$

$= \frac{2n(2n+1)}{2} - \frac{2n(n+1)}{2}$

$= (2n^2 + n) - (n^2 + n)$

$= n^2$

Ex. (4) Find the sum of all odd numbers from 1 to 150.

Solution : 1 to 150 all odd numbers are $1, 3, 5, 7, \ldots, 149$.

Which is an A.P.

Here $a = 1$ and $d = 2$. First let's find how many odd numbers are there from 1 to 150, so find the value of n, if $t_n = 149$

$t_n = a + (n - 1)d$

$149 = 1 + (n - 1)2$ $\quad\therefore\ 149 = 1 + 2n - 2$

$\therefore\ n = 75$

Now let's find the sum of these 75 numbers $1 + 3 + 5 + \ldots + 149$.

$a = 1$ and $d = 2$, $n = 75$

Method I $S_n = \frac{n}{2}[2a + (n-1)d]$

$S_n =$ ☐

$S_n =$ ☐ $\times$ ☐

$S_n =$ ☐

Method II $S_n = \frac{n}{2}[t_1 + t_n]$

$S_n = \frac{75}{2}[1 + 149]$

$S_n =$ ☐ $\times$ ☐

$S_n =$ ☐

Practice Set 3.3

1. First term and common difference of an A.P. are 6 and 3 respectively ; find S_{27}.

$a = 6,\ d = 3,\ S_{27} = ?$

$S_n = \frac{n}{2}\ [\square + (n-1)\ d]$

$S_{27} = \frac{27}{2}\ [12 + (27-1)\ \square]$

$= \frac{27}{2} \times \square$

$= 27 \times 45 \quad = \square$

2. Find the sum of first 123 even natural numbers.

3. Find the sum of all even numbers between 1 and 350.

4. In an A.P. 19th term is 52 and 38th term is 128, find sum of first 56 terms.

5. Complete the following activity to find the sum of natural numbers between 1 and 140 which are divisible by 4.

Between 1 and 140, natural numbers divisible by 4

↓

4, 8, ... , 136

↓

How many numbers ? $\therefore\ n = \square$

↓

$n = \square\ ,\ a = \square\ ,\ d = \square$

↓

$t_n = a+(n-1)d$

↓

$136 = \square + (n - 1) \times \square$

→ $n = \square$ → $S_n = \frac{n}{2}\ [2a+(n-1)d]$

$S_{\square} = \frac{\square}{2}\ [\qquad\qquad] \quad = \square$

Sum of numbers from 1 to 140, which are divisible by 4 = $\square$

6. Sum of first 55 terms in an A.P. is 3300, find its 28th term.